

Feature	Mobile Web	Hybrid App	Native App
User experience	Average	Average	Excellent
Can use device features	No	To some extent	Excellent
Time taken	Less	Less	More
Performance	Average	Average	Best
Portability	Excellent	Good	Bad
App store presence	No	Yes	Yes
Cost of development	Less	Average	More
Technologies	HTML5, CSS3, Twitter Bootstrap	PhoneGap, jQueryMobile, Sencha touch	Objective-C for iPhones and iPad. Android Development Tools (ADT) for the Android OS. Likewise, each platform has its own set of tools and technologies.

websites can't leverage the device's features like the camera, phone-book, etc. The absence of an app store prevents a relatively good user experience.

Hybrid apps

Is it enough to build a website with a responsive Web design? Unfortunately, it is not. Sophisticated smartphone features like the camera, phonebook, etc., are often underutilised. You can't have the app store attendance with the website itself. With a hybrid app one can control all the device features and have the app store presence as well. It is not a true mobile app (native app) but disguised as one. There is no need to build for each and every mobile platform. Fundamentally, these are Web apps wrapped with a thin interface layer to interact with the device platform and to control the device's features (like the camera, phonebook, device settings, etc). PhoneGap and jQuery Mobile are open source frameworks available for building hybrid apps.

jQuery Mobile

As the name suggests, the jQuery Mobile theme revolves around smartphones. You may need to dress your HTML elements to look and act like mobile widgets. The jQuery Mobile framework supplies these kinds of widgets in a ready-to-serve state. jQuery Mobile, along with PhoneGap, appeals a lot to those building hybrid apps.

PhoneGap

The PhoneGap framework takes the 'write less, do more' mantra to the next level. Instead of writing unique apps for each mobile device or OS, the PhoneGap framework allows you to design a single, highly-branded app that will work on all popular smartphone and tablet platforms. PhoneGap is a free and open source framework that allows you to create mobile apps using standardised Web APIs for the platforms you care about. PhoneGap supports all the leading mobile platforms. You can easily create apps using the Web technologies you know and love: *HTML, CSS and JavaScript*.

The pros and cons of hybrid apps

Hybrid apps can be built with less development effort compared to native apps. These apps are able to tap the device's features. And the learning curve for app developers is less steep. These apps are able to ensure a presence in the app store.

The disadvantage is that hybrid apps lag behind native apps slightly with respect to the user experience and performance.

Native apps

Unlike the hybrid apps, you need to develop an app for each and every mobile platform (OS) you intend to support. Native apps provide the best usability, the best features and the best overall mobile experience. The best way to develop highly refined and stylish apps, unless pressed for time and hampered by budget constraints, is the native app way.

The technologies available for this option depend on the mobile OS chosen. For example, to build an iPhone app, you need to develop in objective-C.

The pros and cons of native apps

Native apps ensure the best mobile user experience and app store presence. A lot of support is available to help build these apps, from platform vendors. Native apps leverage the device's features to the maximum extent.

On the downside, since there are many platforms to support, a lot of time and a vast budget are required to develop native apps. It is also difficult to get talented resources for each mobile OS, considering that a new OS shows up every other day. 

References

- [1] <http://phonegap.com/>
- [2] <http://getbootstrap.com/>
- [3] <http://jquerymobile.com/>
- [4] <http://www.sencha.com/>

By: Pradeep Kumar Meruva

The author is a solution architect at Ericsson with over 10 years' experience in open source technologies. After a decade with Java related technologies, he switched over to mobile related technologies, particularly to develop scalable Web applications. You can reach him at pradeep.meruva@gmail.com.